ARTIFICIAL INTELLIGENCE

CSE 3201

Introduction to AI

What is AI?

- **Bellman, 1978**

"[The automation of] activities that we associate with human thinking, activities such as decision making, problem solving, learning"

- **Charniak & McDermott, 1985**

"The study of mental faculties through the use of computational models"

- **Luger & Stubblefield, 1993**

"The branch of computer science that is concerned with the automation of intelligent behavior"

What is AI?

- Dean et al., 1995

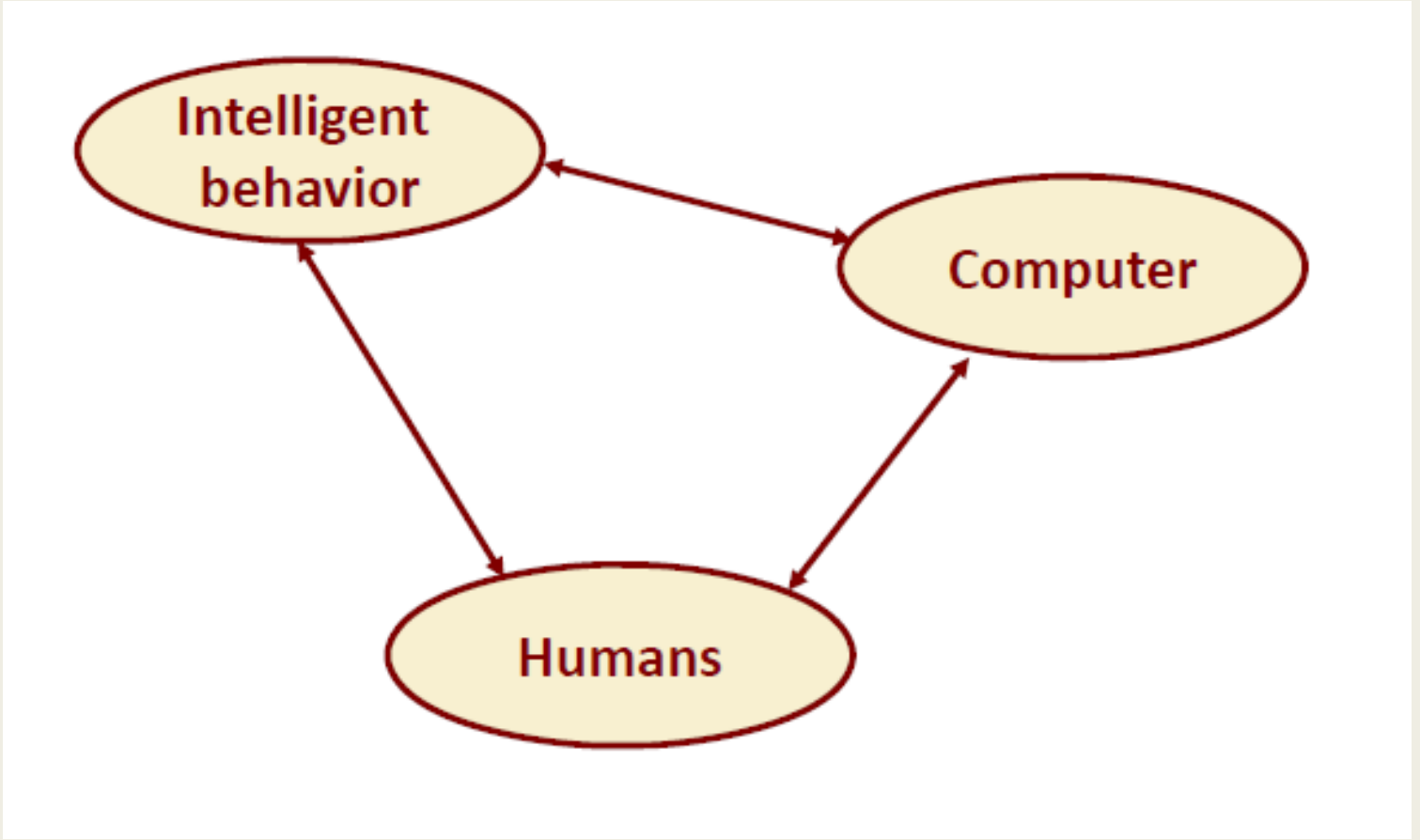
"The design and study of computer programs that behave intelligently. These programs are constructed to perform as would a human or an animal whose behavior we consider intelligent"

- Rich & Knight, 1991

"The study of how to make computers do things at which, at the moment, people are better"

- **Examples:** Speech recognition, Face, Object, Inferencing, Learning new skills, Decision making, translations between languages etc.

What is AI?



What is Intelligence?

- “ability to learn, understand and think” (Oxford dictionary)
- “The capacity to learn and solve problems” [Webster dictionary]
- Ability to think and act rationally.

So, What are “ingredients” of intelligence?

“Ingredients” of Intelligence

- Ability to interact with real world
  - Perceive, understand, act.
  - Language understanding and formation.
  - Visual perception.

“Ingredients” of Intelligence

- Reasoning and Planning
  - Problem solving, planning, decision making
  - Ability to deal with unexpected problems, dealing with uncertainty

- Learning and Adaptation
  - Continuous update of our model of the world and adaptation to it

Weak AI and Strong AI

- **Weak AI:**

- The principle behind Weak AI is simply the fact that machines can be made to act as if they are intelligent.

- For example, when a human player plays chess against a computer, the human player may feel as if the computer is actually making impressive moves. But the chess application is not thinking and planning at all. All the moves it makes are previously fed in to the computer by a human and that is how it is ensured that the software will make the right moves at the right times.

Weak AI and Strong AI

- **Strong AI:**

- The principle behind Strong AI is that the machines could be made to think or in other words could represent human minds in the future.
- As its name shows it is an advanced AI system which has the capability to perform the task on its own without the external help. Those machines will have the ability to think and do all functions that a human is capable of doing.
- Currently, we are in the initial stage of development of strong AI. So it's a long way to do such tasks which the strong AI is designed for but research is being carried out on it.

Weak AI vs Strong AI

- In term of weak AI, it has the ability to perform an only specific set of tasks. For example, play chess game, so here if you come here to taking with your other simple problem then due to the reason that this problem is outside of its capability, which it is designed to work(Play Chess), So you would not going to get any response from it.

- If we come with strong AI then it has the capability of problem-solving like the human being. It does not mean that it can solve any type of problems, any individual human can’t even do that thing, but they have the capability to solve the wide range of problems.

Weak AI vs Strong AI

- Another key difference is between:

  “we programmed it to solve these things” (Weak AI) and

  “it has learned to solve these things”(Strong AI)

- Machines can be made to behave as if they were intelligent (Weak AI)

  Machines can have consciousness (Strong AI)

Why AI?

- Human brain learn through series of experiences or repeated events to similar things. A child brain doesn't know what an apple is or how to distinguish between a tiger and a leopard. But when his brain is been fed with knowledge about similar instances, THE BRAIN never forgets so easily. Now, one never need to look at his notes to know that "ok, it is round and red, It may be an apple."

- Keeping the idea alive, scientists thought to train modern computers to think like a brain. Obviously, that cannot be done without repeated training to computers. We can say, they tried to provide artificial brain to the machines. Major AI researchers and textbooks define this field as "the study and design of intelligent agents", where an **intelligent agent** is a system that perceives its environment and takes actions that maximize its chances of success.

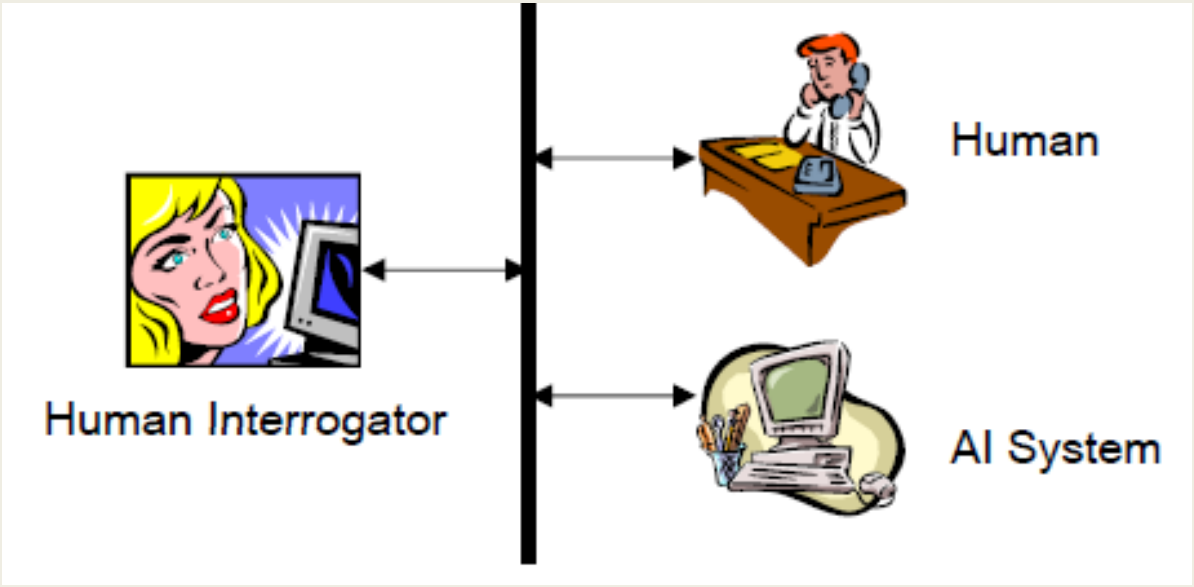
Learning artificial intelligence field could help you to perform complex tasks that are otherwise difficult to perform.

AI Characterizations

	HUMAN	RATIONAL
THOUGHT	Systems that think like humans	Systems that think rationally
BEHAVIOUR	Systems that act like humans	Systems that act rationally

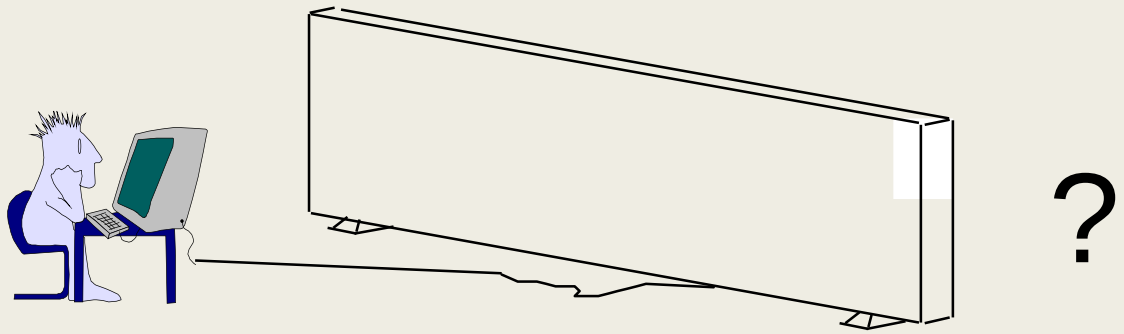
Acting Humanly: The Turing Test

- Alan Turing (1912-1954)
- "The art of creating machines that perform functions that require intelligence when performed by people." (Kurzweil)
- Interrogator interacts with a computer and a person via a teletype. (remote communication)
- Computer passes the Turing test if interrogator cannot determine which is which.



Acting Humanly: The Turing Test

- You enter a room which has a computer terminal. You have a fixed period of time to type what you want into the terminal, and study the replies. At the other end of the line is either a human being or a computer system.



?

- If it is a computer system, and at the end of the period you cannot reliably determine whether it is a system or a human, then the system is deemed to be intelligent.

Acting Humanly: The Turing Test

- These tasks include:
  - *Natural language processing*
    - for communication with human
  - *Knowledge representation*
    - to store information efficiently
  - *Automated reasoning*
    - to retrieve & answer questions using the stored information
  - *Machine learning*
    - to adapt to new circumstances

Total Turing Test

- Includes two more issues: (So that the interrogator can test the subject's perceptual abilities)
  - *Computer Vision*
    - to perceive objects (seeing)
  - *Robotics*
    - to move objects (acting)

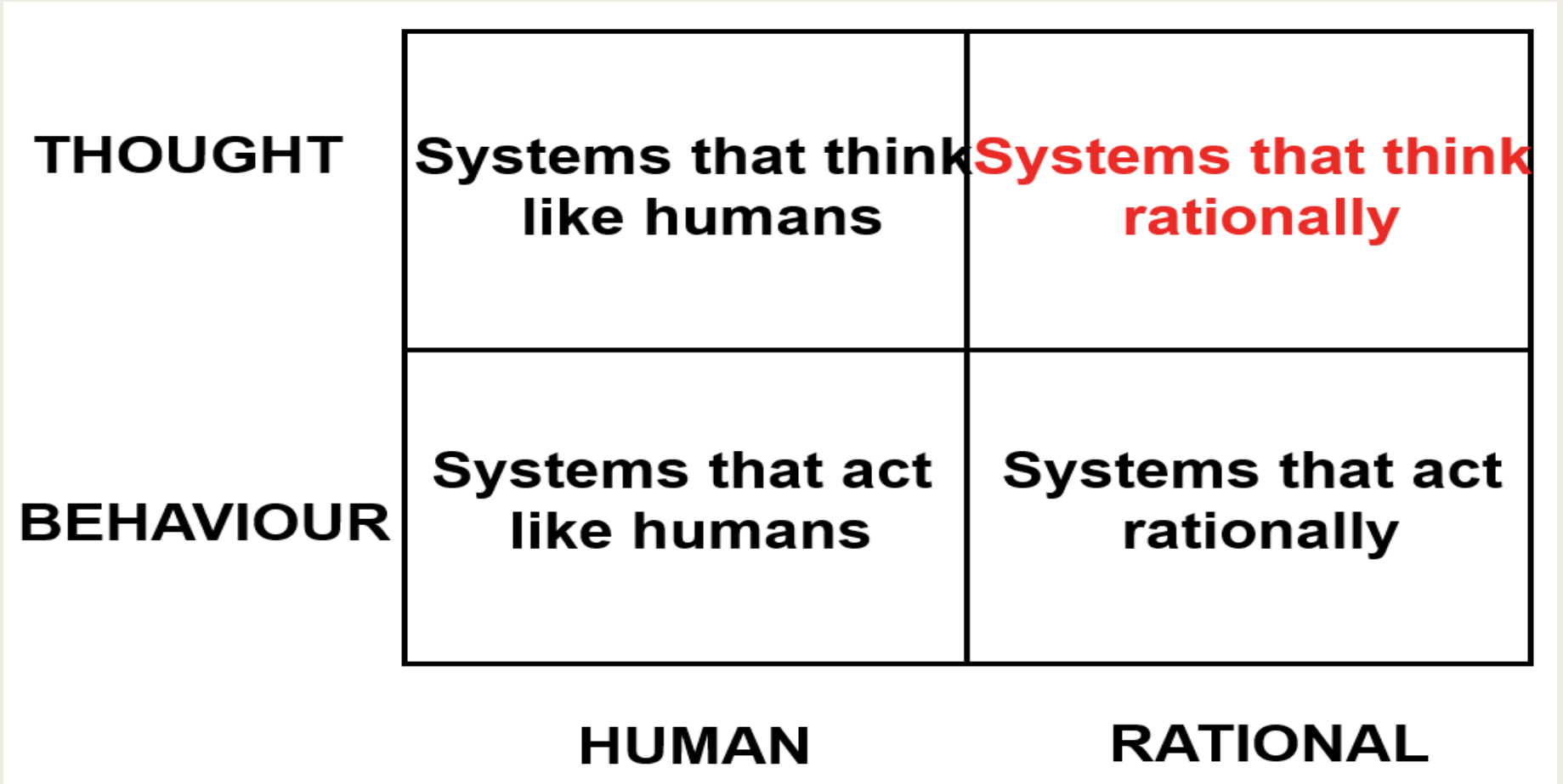
AI Characterizations

	HUMAN	RATIONAL
THOUGHT	Systems that think like humans	Systems that think rationally
BEHAVIOUR	Systems that act like humans	Systems that act rationally

Acting Rationally

- Rational behavior: doing the right thing
- The right thing: that which is expected to maximize goal achievement, given the available information.
- Giving answers to questions is ‘acting’.
- The intelligent agent approach
  - An *agent* is something that perceives and acts
- I don't care whether a system:
  - *replicates human thought processes*
  - *makes the same decisions as human*

AI Characterizations



Thinking Rationally

- *The laws of thought approach – make logical thoughts*
  - "Right Thinking" - Thinking rightly in a given circumstances with given information.
- *Requires 100% knowledge*
  - Need to gather 100% knowledge of a particular problem in the knowledge base to answer rightly every time. But what is 100% knowledge? Challenging indeed.
- *Requires too many computations*
  - Sometimes too many comparisons and computations performed to take a little decision. Thus computational cost increases.

Thinking Humanly

- Thinking humanly means trying to understand and model how the human mind works. **Thinking humanly** is related to cognitive science. It is the theory of how the brain works. How it handles the nodes and how it analyzes the data that your body generates. So far, there is no full understanding on how the brain functions.

- There are (at least) two possible routes that humans use to find the answer to a question:
  - We reason about it to find the answer. This is called “introspection”. (self-knowledge)
  - We conduct experiments to find the answer, drawing upon scientific techniques to conduct controlled experiments and measure change.

Foundations of AI

- **Cognitive Science:**

- As a way to understand how natural mind works. e.g., visual perception, memory, learning etc.
- **Cognitive science** is the study of the mind, including its structure and everything it does. It includes a variety of research sciences, including Education, Philosophy, Psychology, Neuroscience, Linguistics etc.
- Cognitive scientists study intelligence and behavior, with a focus on how nervous systems represent, process, and transform information.

[Note: **Cognition** is "the mental action or process of acquiring knowledge and understanding through thought, experience, and the senses“]

Foundations of AI

- **Philosophy:**

– The study of knowledge, reality, and existence.

– As a way to explore some basic and interesting (and important) philosophical questions.

- **Engineering:**

– To get machines to do a wider variety of useful things

e.g., understand spoken natural language, recognize individual people in visual scenes, find the best travel plan for your vacation etc.

*See book Russel and Norvig (3rd Edition) : 1.2

A Brief History of AI

- The gestation of AI (1943 - 1956):

- 1943: McCulloch & Pitts: Boolean circuit model of brain.

- 1950: Turing's "Computing Machinery and Intelligence".

- 1956: McCarthy's name "Artificial Intelligence" adopted.

- Early enthusiasm, great expectations (1952 - 1969):

- Early successful AI programs: Samuel's checkers, Newell Simon's Logic Theorist, Gelernter's Geometry Theorem Prover.

- Robinson's complete algorithm for logical reasoning.

A Brief History of AI

- A dose of reality (1966 - 1974):

- AI discovered computational complexity.

- Neural network research almost disappeared after Minsky & Papert's book in 1969.

- Knowledge-based systems (1969 - 1979):

- 1969: DENDRAL by Buchanan et al..

- 1976: MYCIN by Shortliffle.

- 1979: PROSPECTOR by Duda et al..

A Brief History of AI

- AI becomes an industry (1980 - 1988):

- Expert systems industry booms.

- 1981: Japan's 10-year Fifth Generation project.

- The return of NNs and novel AI (1986 - present):

-Mid 80's: Back-propagation learning algorithm reinvented.

- Expert systems industry busts.

- 1988: Resurgence of probability.

- 1988: Novel AI (ALife, GAs, Soft Computing, …).

- 1995: Agents everywhere.

- 2003: Human-level AI back on the agenda.

Branches of AI/AI Subsets

